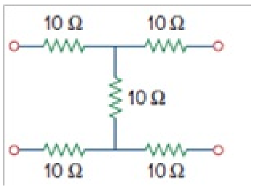
Chapter 19, Problem 28P

Problem

In the circuit of Fig. 19.65, the input port is connected to a 1−A current source and the right hand side of the circuit is left open ($I2 = 0$). Calculate the power absorbed by the circuit by using the *y* parameters. Confirm your result by direct circuit analysis.

Figure 19.65



Step-by-step solution

Step 1 of 20

Refer to Figure 19.65 from the textbook.

The admittance parameters of the network are given by the two equation as follows:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$I_1 = y_{11}V_1 + y_{12}V_2$ …… (1)

$I_2 = y_{21}V_1 + y_{22}V_2$ …… (2)

Here,

V_1 is the voltage across input port,

V_2 is the voltage across output port,

I_1 is the current entering the input port,

I_2 is the current entering the output port.

The admittance parameters are:

$$y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}$$

$$y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$

$$y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$$

$$y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

Step 2 of 20

Assume $V_2 = 0$. Sketch the circuit as shown in Figure 1.

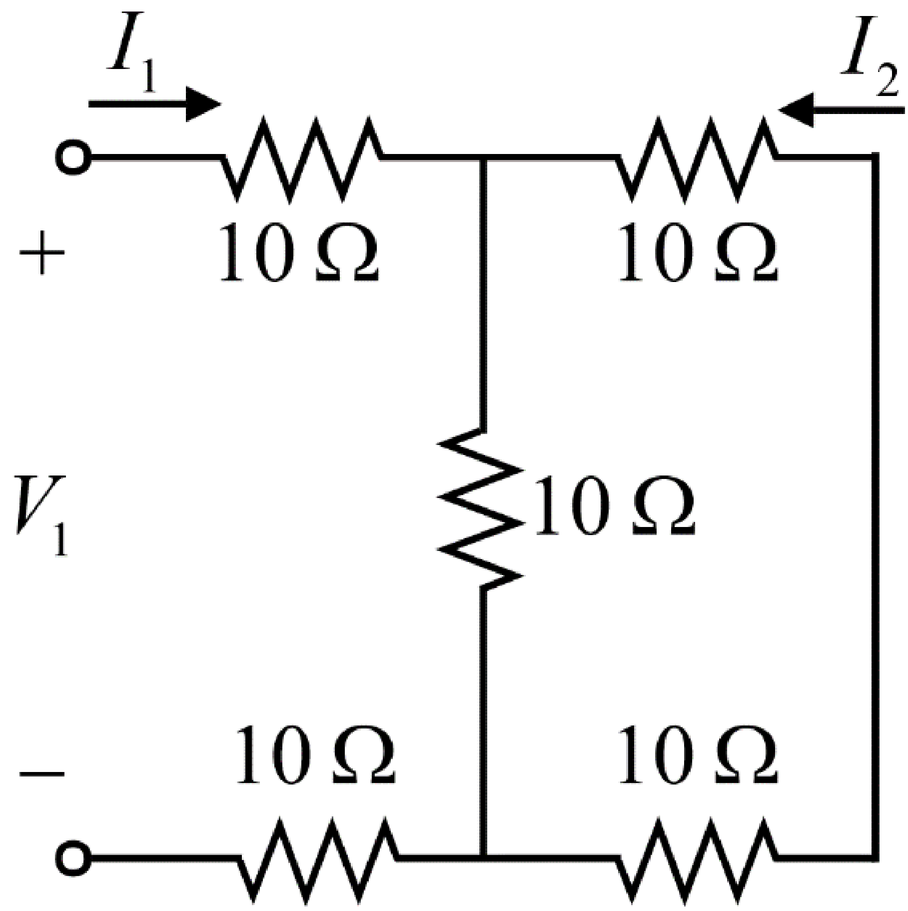


Figure 1

Step 3 of 20

The two $10\,\Omega$ resistors are in series.

Calculate the series equivalent of the two resistors.

$$\begin{aligned} R_{s1} &= 10+10 \\ &= 20\,\Omega \end{aligned}$$

Simplify the circuit as shown in Figure 2.

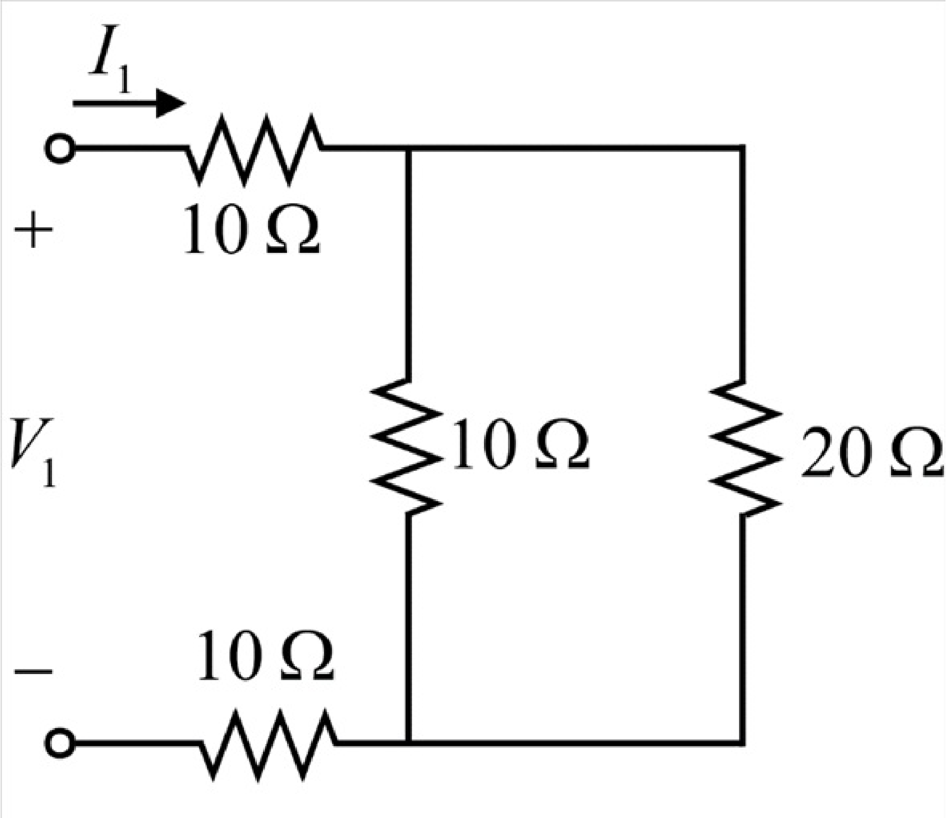


Figure 2

Step 4 of 20

The $20\,\Omega$ resistor is in parallel with the $10\,\Omega$ resistor.

Calculate the parallel equivalent of the two resistors.

$$\begin{aligned} R_{p1} &= \frac{(10)(20)}{10+20} \\ &= \frac{20}{3}\,\Omega \end{aligned}$$

Step 5 of 20

Sektch the simplified circuit as shown in Figure 3.

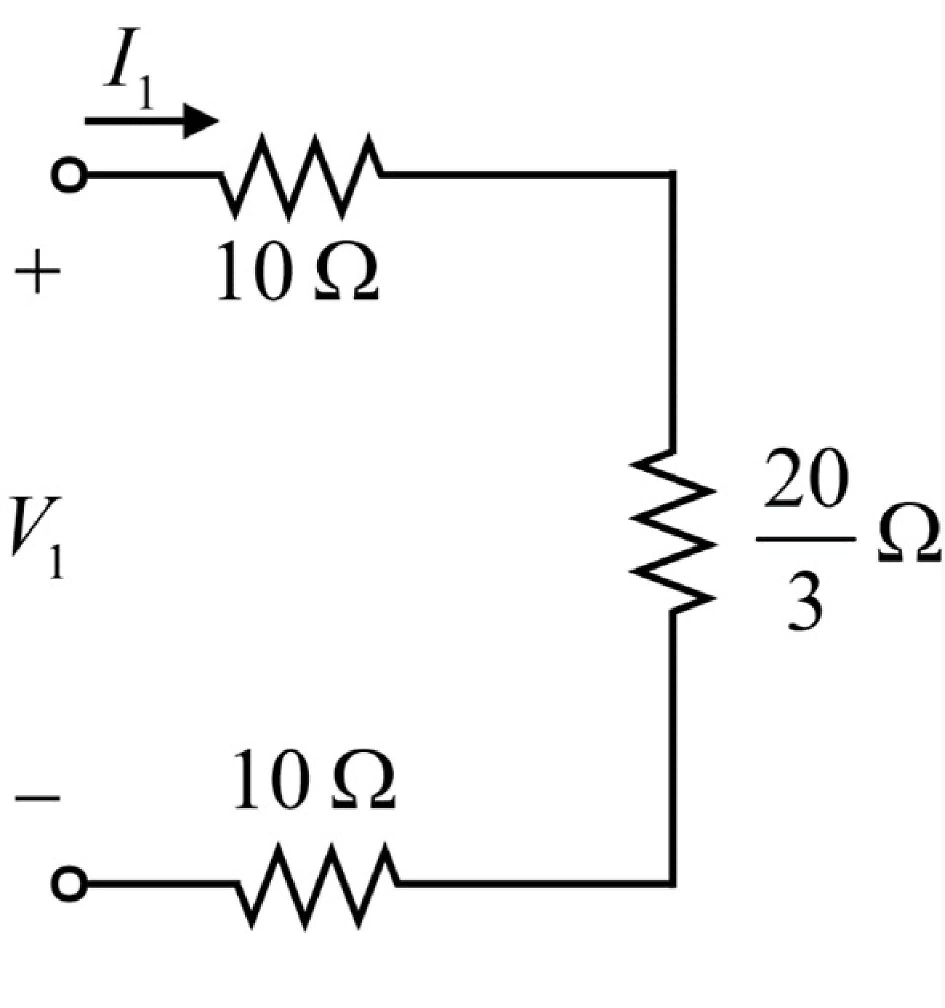


Figure 3

Step 6 of 20

Apply Kirchhoff's Voltage Law in the circuit shown in Figure 3.

$$V_1 = I_1\left(10+10+\frac{20}{3}\right)$$
$$V_1 = \frac{80}{3}I_1$$
$$\frac{I_1}{V_1} = \frac{3}{80}$$

Since $y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}$.

Thus,

$$\begin{aligned} y_{11} &= \frac{3}{80} \\ &= 0.0375\ \mathrm{S} \end{aligned}$$

Step 7 of 20

Apply Current Division Rule in Figure 2 to calculate I_2.

$$I_2 = -\left(\frac{10}{10+20}\right)I_1$$

Substitute $\frac{3}{80}V_1$ for I_1.

$$I_2 = -\left(\frac{10}{10+20}\right)\left(\frac{3}{80}V_1\right)$$
$$I_2 = -\left(\frac{1}{3}\right)\left(\frac{3}{80}V_1\right)$$
$$I_2 = -\frac{1}{80}V_1$$
$$\frac{I_2}{V_1} = -\frac{1}{80}$$

Since $y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$.

Thus,

$$\begin{aligned} y_{21} &= -\frac{1}{80} \\ &= -0.0125\text{ S} \end{aligned}.$$

Step 8 of 20

Assume $V_1 = 0$. Sketch the circuit as shown in Figure 4.

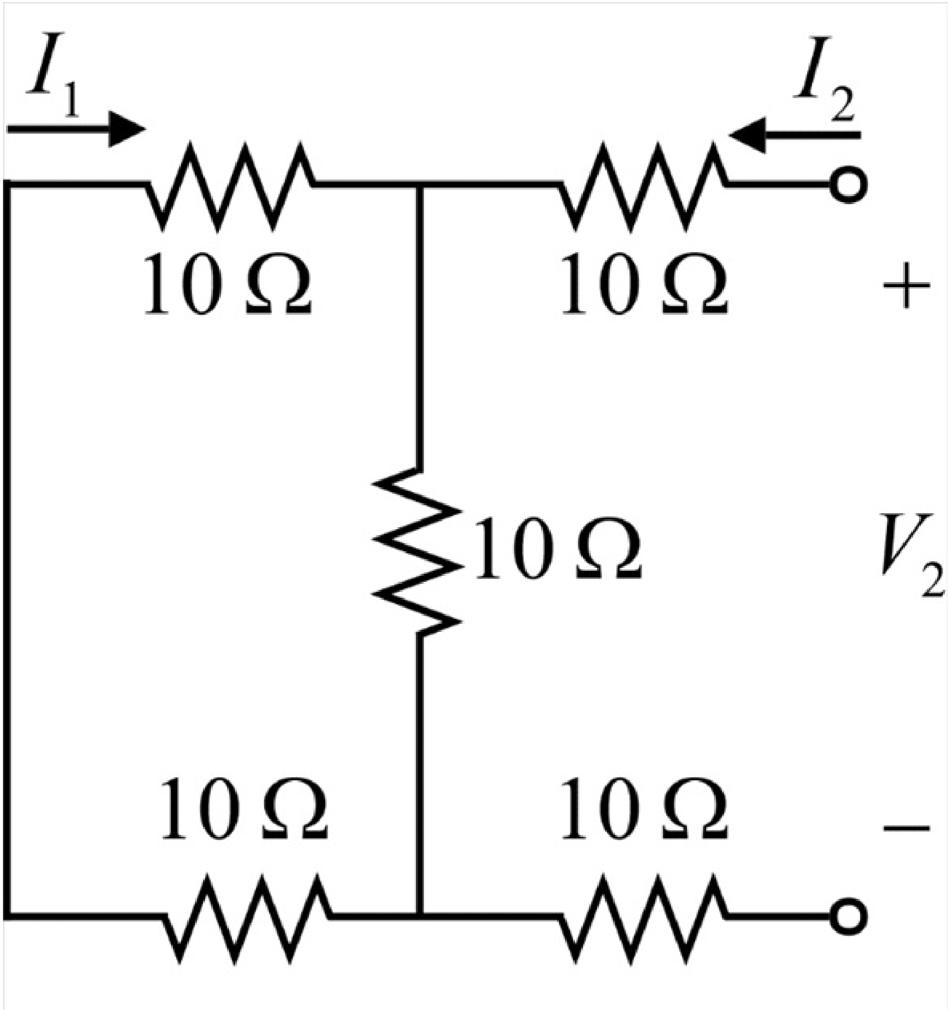


Figure 4

The two $10\,\Omega$ resistors are in series.

Calculate the series equivalent of the two resistors.

$$\begin{aligned} R_{s2} &= 10+10 \\ &= 20\,\Omega \end{aligned}$$

Simplify the circuit as shown in Figure 5.

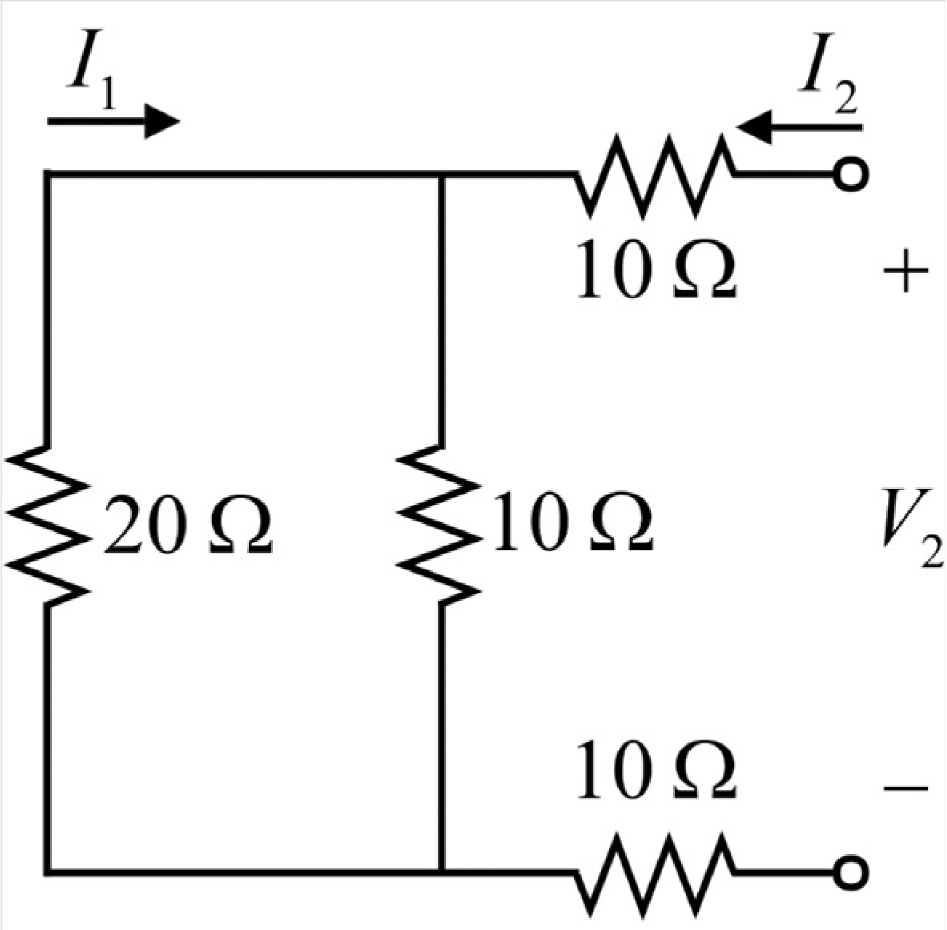


Figure 5

The $20\,\Omega$ resistor is in parallel with the $10\,\Omega$ resistor.

Calculate the parallel equivalent of the two resistors.

$$R_{p2} = \frac{(10)(20)}{10+20}$$
$$= \frac{20}{3}\,\Omega$$

Sektch the simplified circuit as shown in Figure 6.

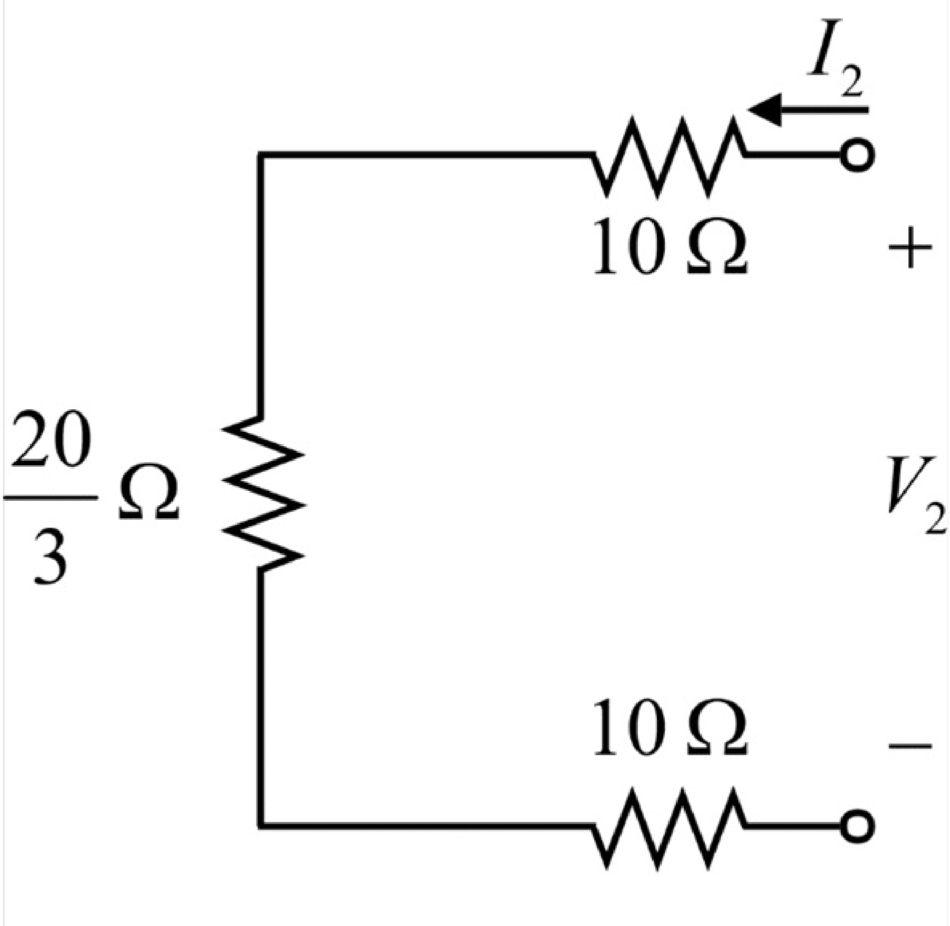


Figure 6

Step 12 of 20

Apply Kirchhoff's Voltage Law in the circuit shown in Figure 3.

$$V_2 = I_2\left(10+10+\frac{20}{3}\right)$$
$$V_2 = \frac{80}{3}I_2$$
$$\frac{I_2}{V_2} = \frac{3}{80}$$

Since $y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$.

Thus,

$$y_{22} = \frac{3}{80}$$
$$= 0.0375\text{ S}$$

Step 13 of 20

Apply Current Division Rule in Figure 5 to calculate I_1.

$$I_1 = -\left(\frac{10}{10+20}\right) I_1$$

Substitute $\frac{3}{80}V_2$ for I_2.

$$I_1 = -\left(\frac{10}{10+20}\right)\left(\frac{3}{80}V_2\right)$$
$$I_1 = -\left(\frac{1}{3}\right)\left(\frac{3}{80}V_2\right)$$
$$I_1 = -\frac{1}{80}V_2$$
$$\frac{I_1}{V_2} = -\frac{1}{80}$$

Since $y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$.

Thus,

$$\begin{aligned} y_{12} &= -\frac{1}{80} \\ &= -0.0125\ \text{S} \end{aligned}$$

Step 14 of 20

Write the y parameter matrix, $[y]$.

$$[y] = \begin{bmatrix} 0.0375 & -0.0125 \\ -0.0125 & 0.0375 \end{bmatrix} \text{S}$$

Step 15 of 20

Substitute $1\ \text{A}$ for I_1, 0 for I_2, $0.0375\ \text{S}$ for y_{11} and $-0.0125\ \text{S}$ for y_{12} in equation (1).

$1 = 0.0375V_1 - 0.0125V_2$ …… (3)

Substitute $1\ \text{A}$ for I_1, 0 for I_2, $0.0375\ \text{S}$ for y_{22} and $-0.0125\ \text{S}$ for y_{21} in equation (2).

$$0 = -0.0125V_1 + 0.0375V_2$$
$$V_2 = \frac{0.0125}{0.0375}V_1$$

Step 16 of 20

Substitute $\frac{0.0125}{0.0375}V_1$ for V_2 in equation (3).

$$1 = 0.0375V_1 - 0.0125\left(\frac{0.0125}{0.0375}V_1\right)$$
$$0.0375 = \left((0.0375)^2 - (0.0125)^2\right)V_1$$
$$V_1 = \frac{0.0375}{0.140625\times10^{-4} - 01.5625\times10^{-4}}$$
$$V_1 = 30\ \text{V}$$

Step 17 of 20

Since the power delivered to the circuit is equal to the power absorbed by it.

Thus,

The power absorbed by the circuit, p is,

$$p = V_1 I_1$$

Substitute $30\ \text{V}$ for V_1 and $1\ \text{A}$ for I_1.

$$\begin{aligned} p &= (30)(1) \\ &= 30\ \text{W} \end{aligned}$$

Step 18 of 20

Direct Analysis:

Since the current, I_2 is 0. Thus, reduce the circuit as shown in Figure 7.

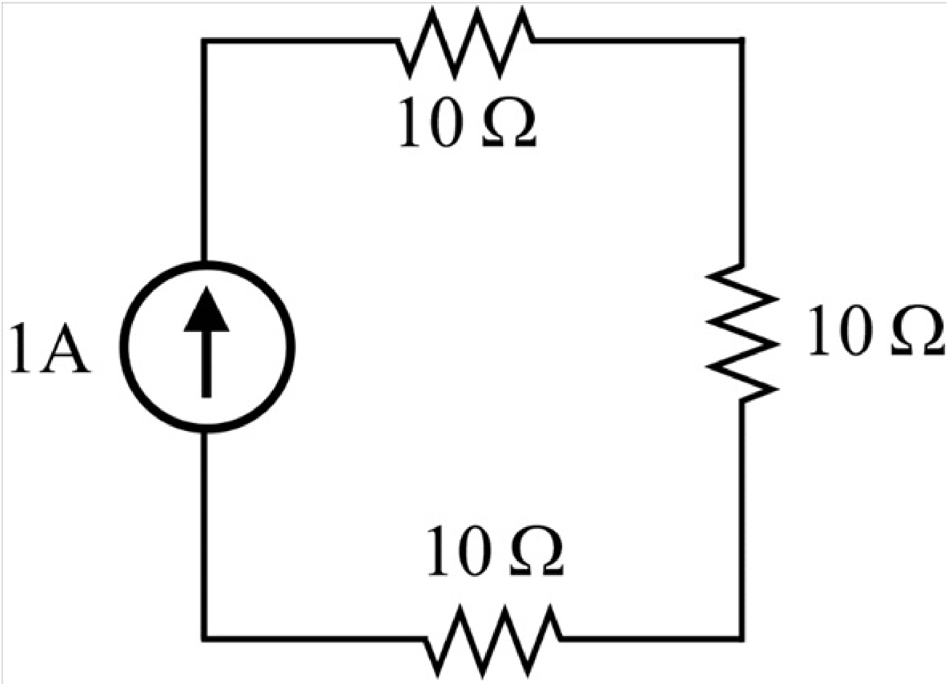


Figure 7

Step 19 of 20

Calulate the series equivalent of the three resistors, R_{eq}.

$$\begin{aligned} R_{eq} &= 10 + 10 + 10 \\ &= 30\ \Omega \end{aligned}$$

Step 20 of 20

Calculate the power absorbed by the circuit, p.

$$p = I^2 R_{eq}$$

Substitute $1\ \text{A}$ for I and $30\ \Omega$ for R_{eq}.

$$\begin{aligned} p &= (1)^2\, 30 \\ &= 30\ \text{W} \end{aligned}$$

Therefore, the power absorbed in the circuit, p is $\boxed{30\ \text{W}}$.